

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-30

Faculty

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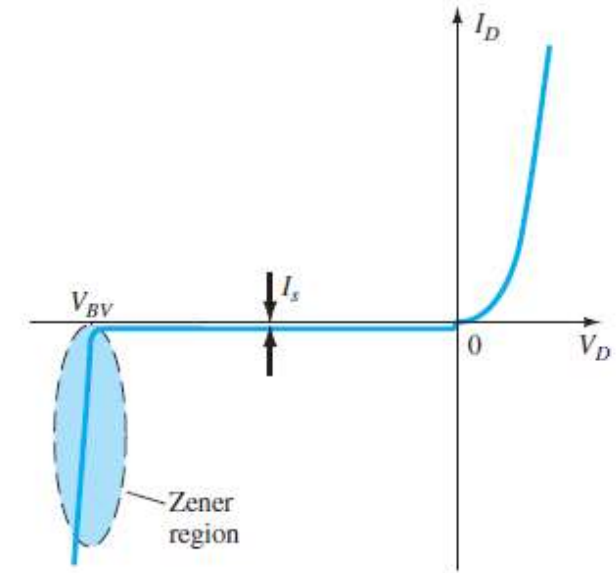
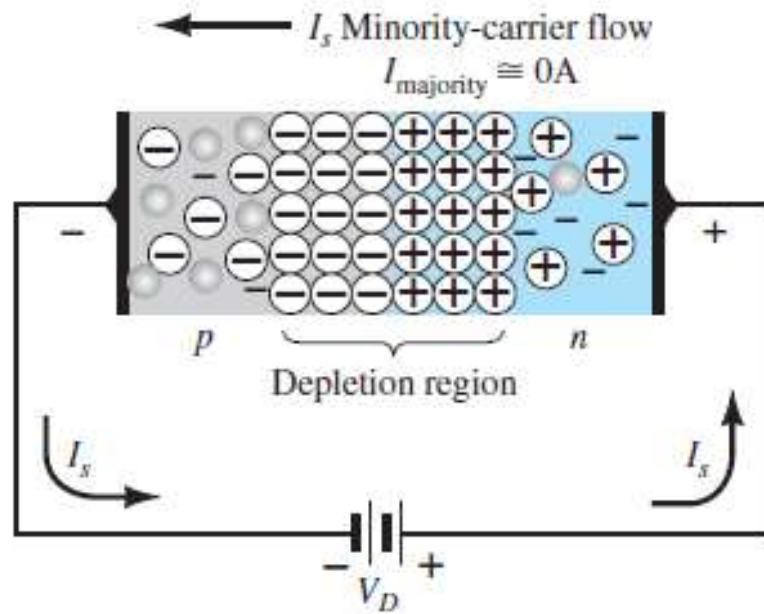
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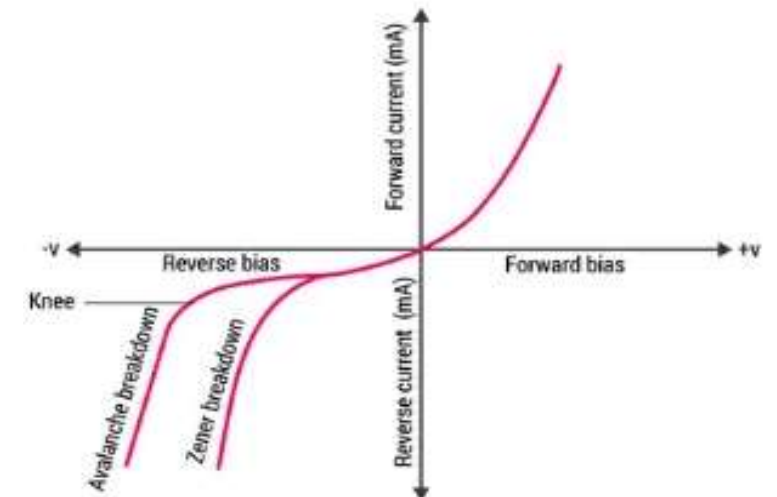
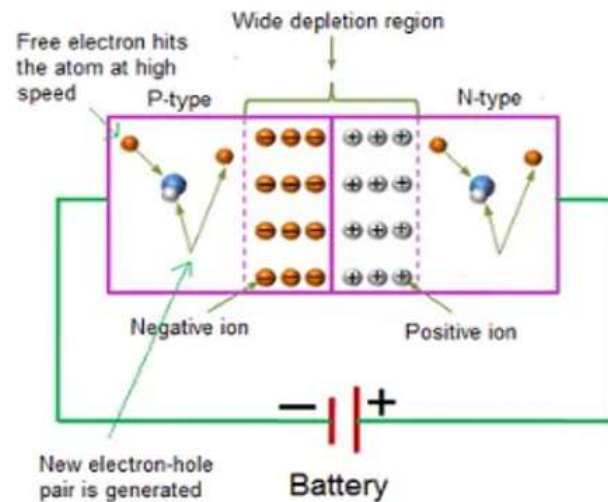
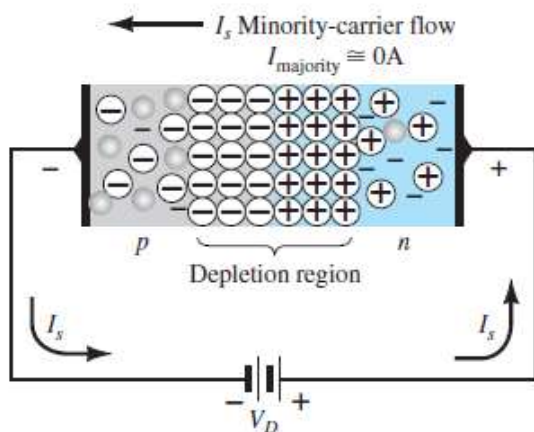
Breakdown Region(Reverse Bias)



1. *avalanche breakdown*
2. *Zener breakdown*

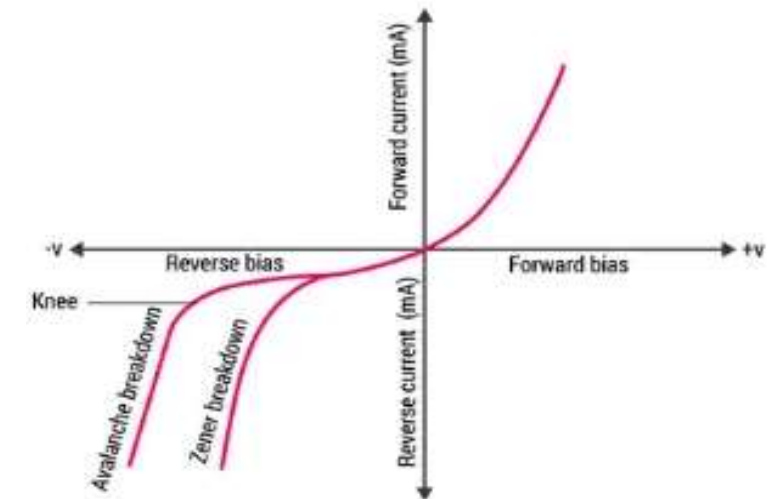
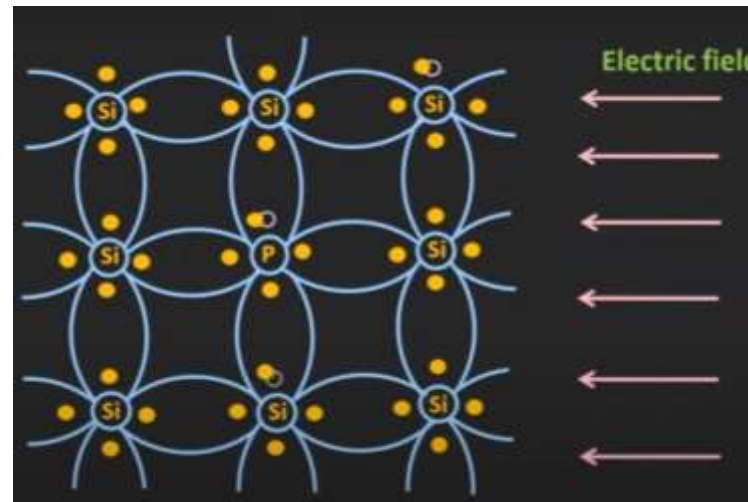
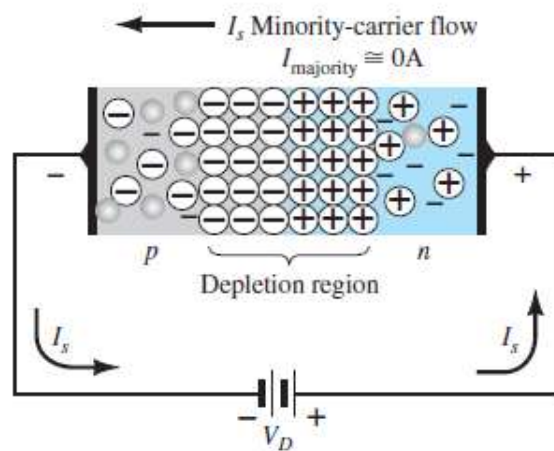
1. Avalanche Breakdown

- As the magnitude of the reverse voltage increases, the kinetic energy of the minority charge carriers also increase. These fast moving electrons collide with the other [atoms](#) in the device to knock-off some more electrons from them.
- The electrons so released further release much more electrons from the atoms by [breaking](#) the covalent bonds. This process is termed as carrier multiplication and leads to a considerable increase in the flow of current through the [p-n junction](#).
- The associated phenomenon is called **Avalanche Breakdown** and the corresponding voltage is Avalanche Breakdown Voltage (V_{BR}), which is a central phenomenon to the working principle of [avalanche diodes](#).

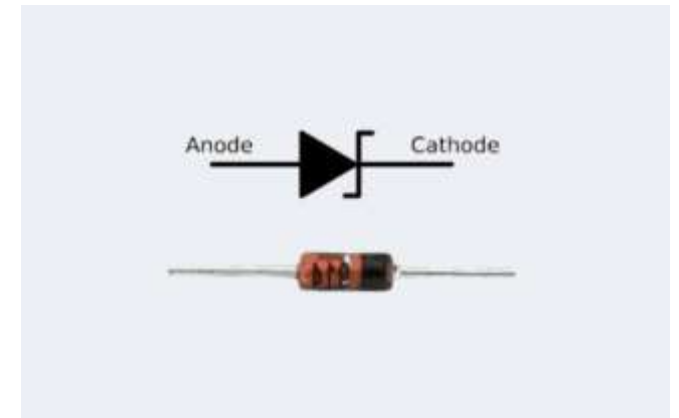
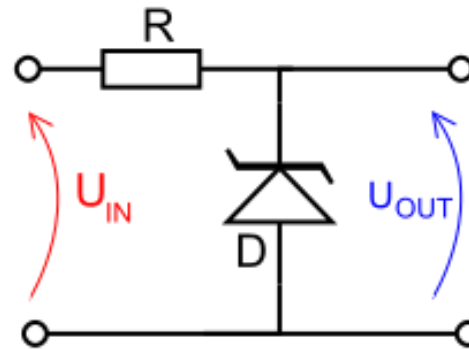
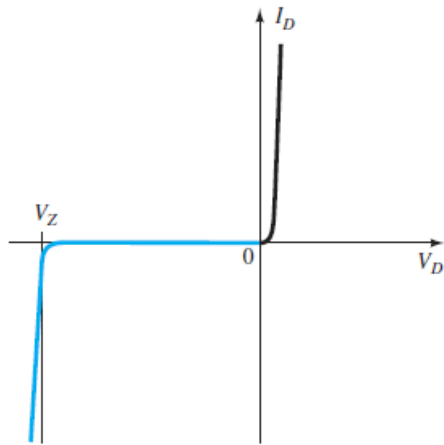
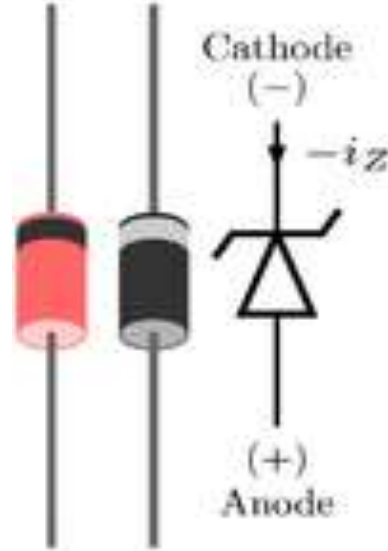
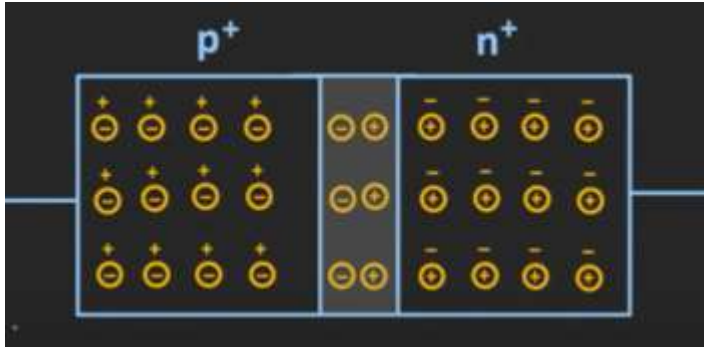


2. Zener Breakdown

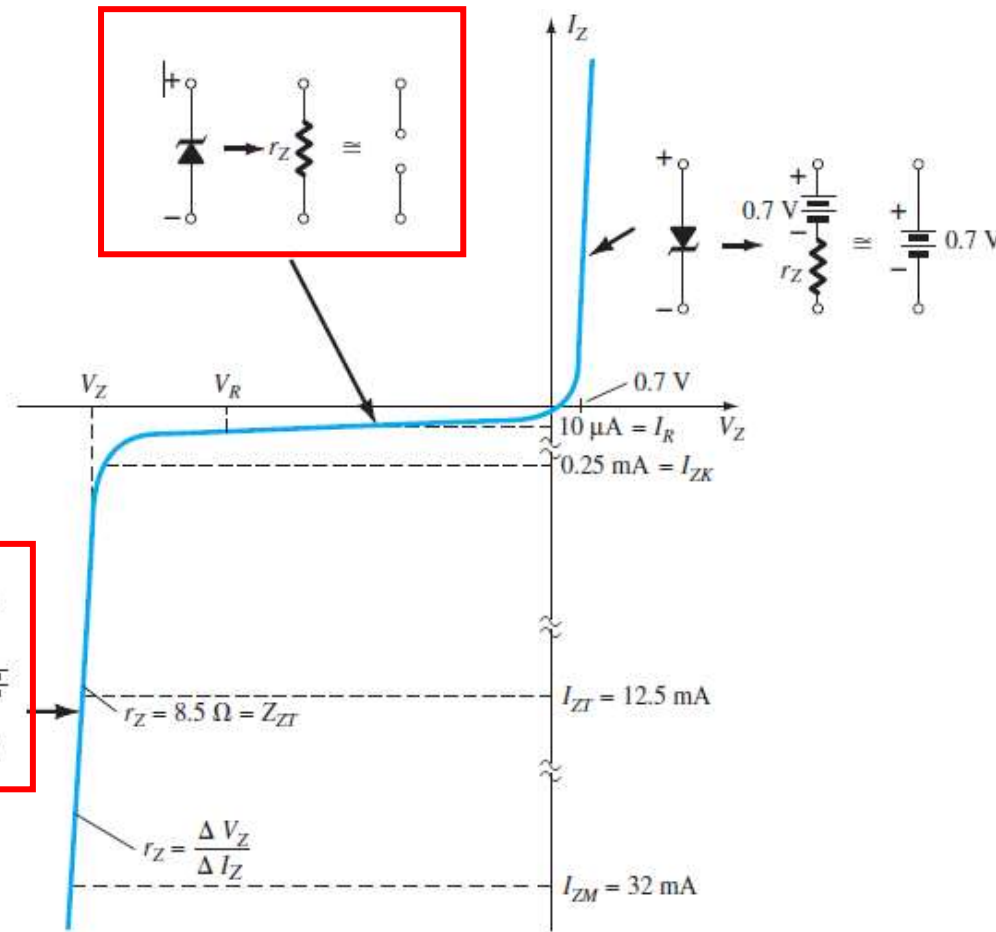
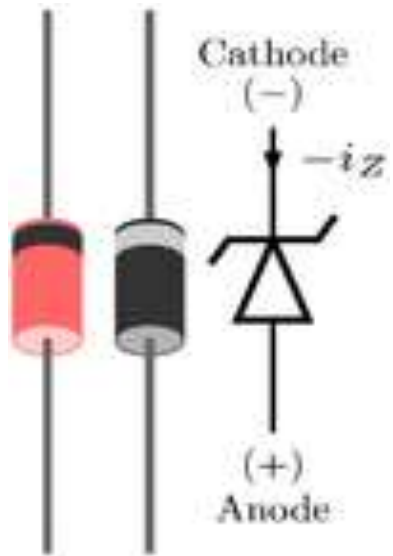
- Under a high reverse-bias voltage, the p-n junction's depletion region widens which leads to a high-strength electric field across the junction.
- Sufficiently strong electric fields enable tunneling of electrons across the depletion region of a semiconductor, leading to numerous free charge carriers. This sudden generation of carriers rapidly increases the reverse current and gives rise to the high slope conductance of the Zener diode.
- The device is named after American physicist Clarence Zener who first described the Zener effect in 1934.



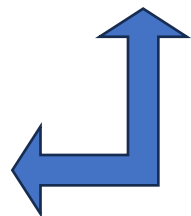
Zener Diodes

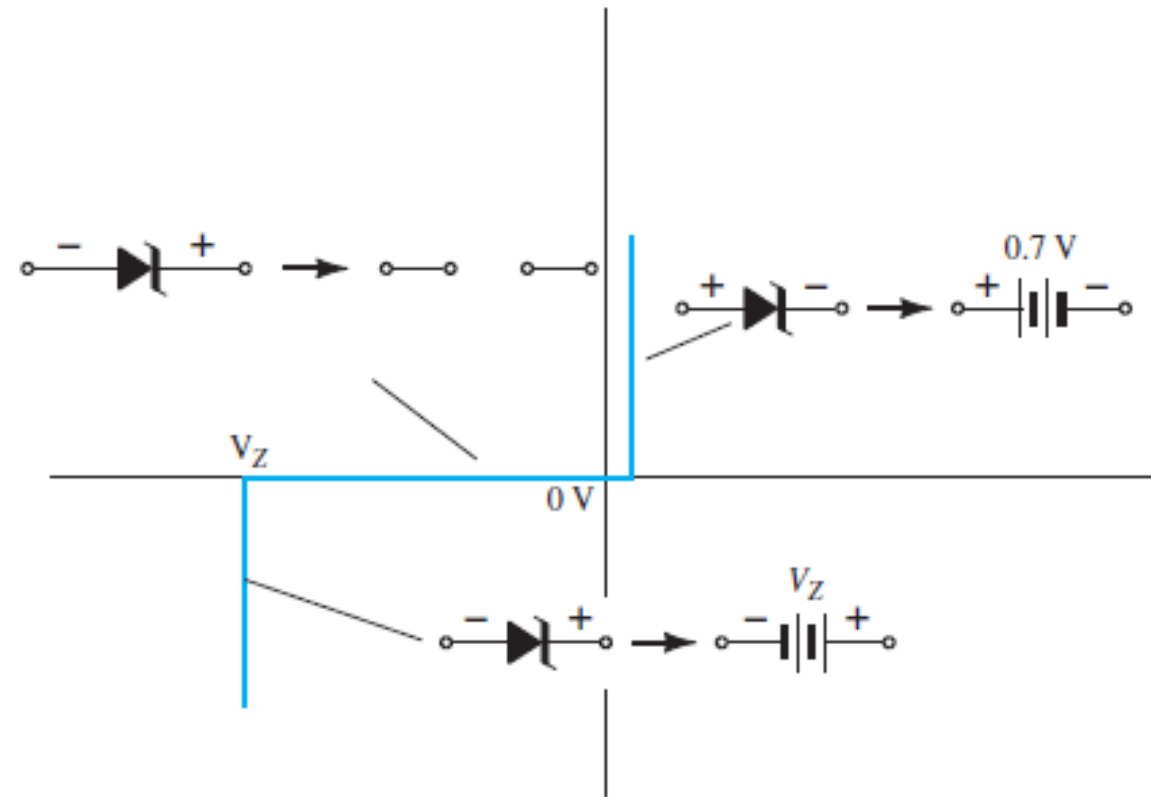


Zener diode characteristics with the equivalent model for each region.



Zener Regulator



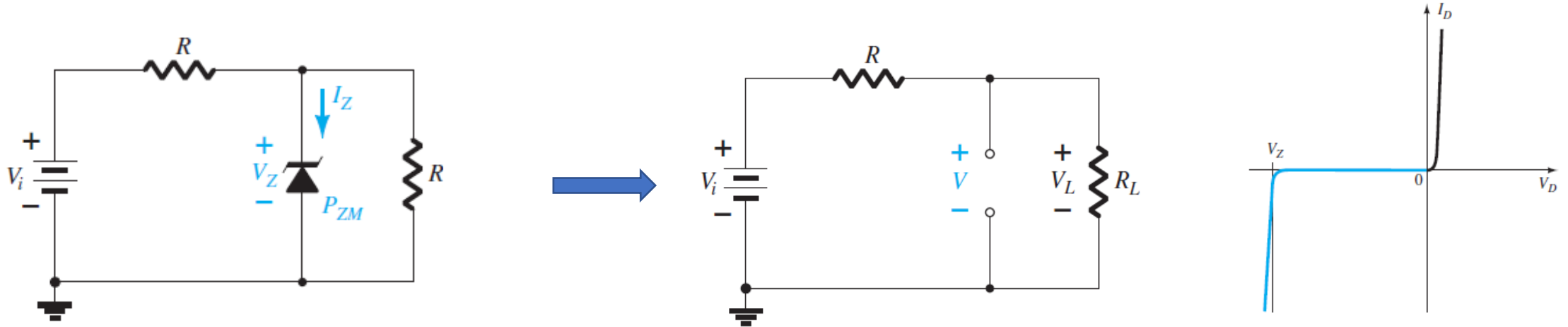


Model Problems:

Case 1: Zener Regulator (Fixed R_L)

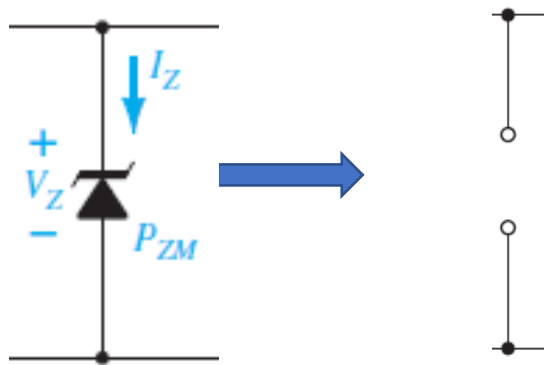
Case 2: Zener Regulator (Variable R_L)

Case 1: Zener Regulator (Fixed R_L)

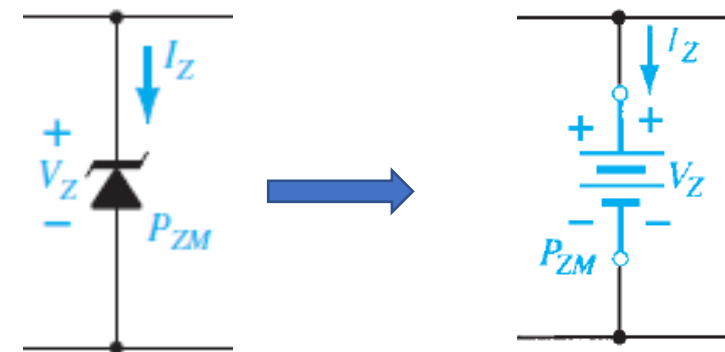


Case-1:

If $V < V_Z$, the diode is “off”



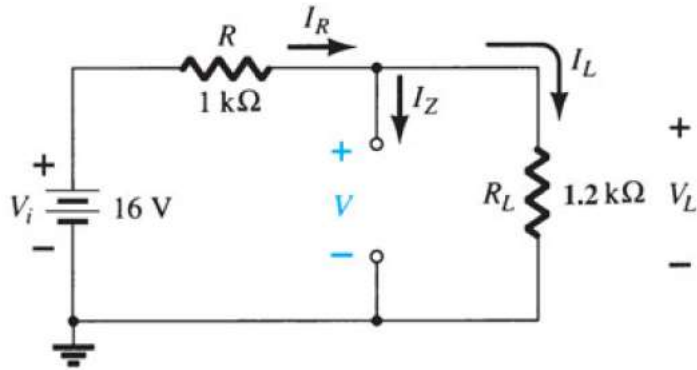
Case-2: If $V \geq V_Z$, the Zener diode is “on”



- The power dissipated by the Zener diode is determined by $P_z = V_z I_z$

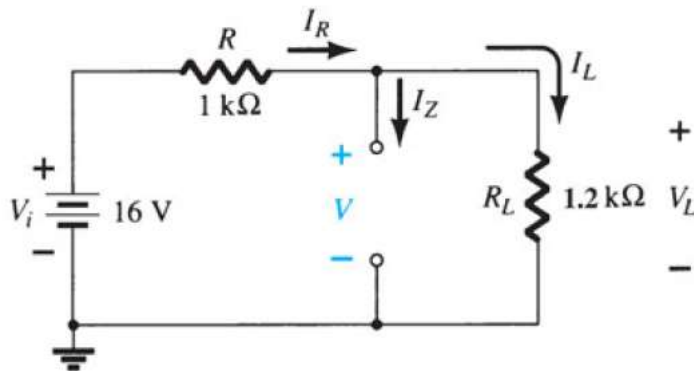
Problem:1

- For the Zener diode network of Fig. below , determine V_L , V_R , I_Z , and P_Z .
- Repeat part (a) with $R_L = 3 \text{ k}\Omega$.



$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$

As $V < V_Z$ then diode is open circuit

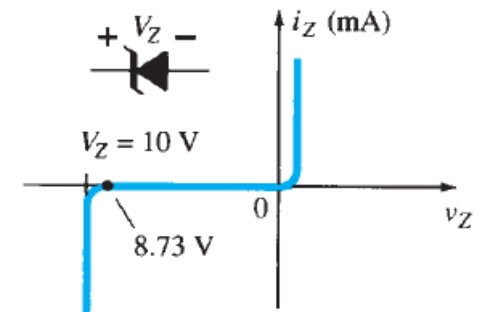
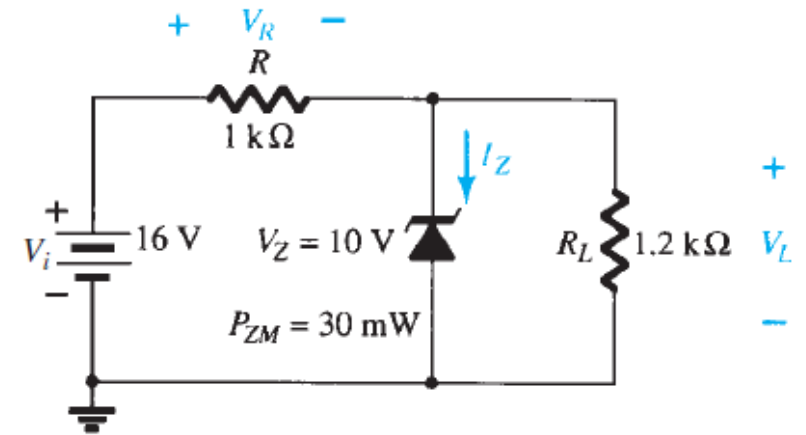


$$V_L = V = 8.73 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

$$I_Z = 0 \text{ A}$$

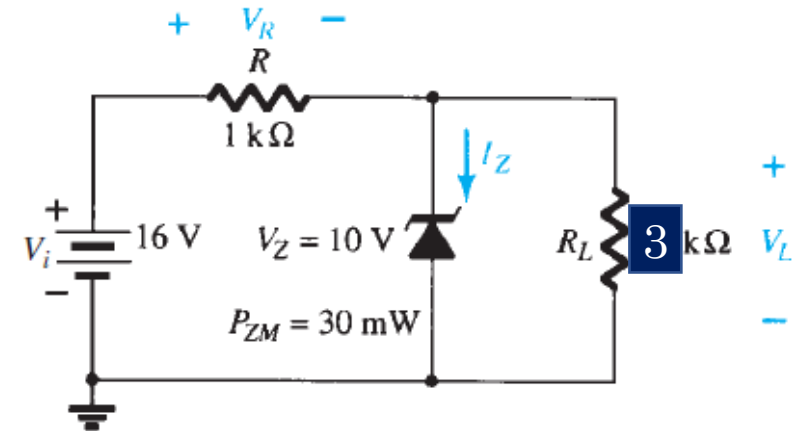
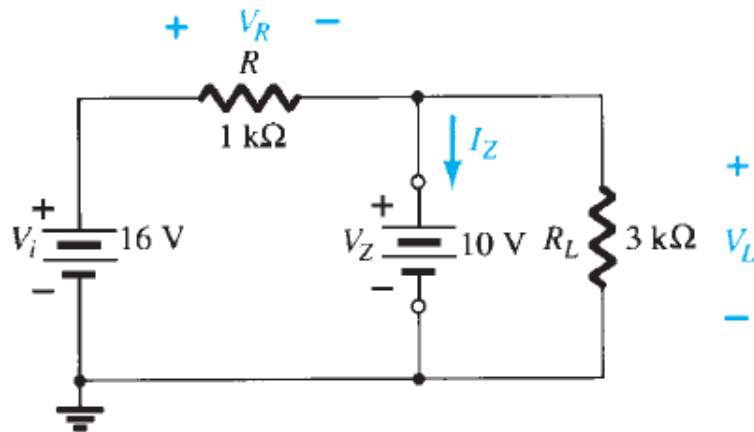
$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$



b) For the Zener diode network of Fig. below, determine V_L , V_R , I_Z , and P_Z .

$$V = \frac{R_L V_i}{R + R_L} = \frac{3 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 3 \text{ k}\Omega} = 12 \text{ V}$$

Since $V = 12 \text{ V}$ is greater than $V_Z = 10 \text{ V}$, the diode is in the “on” state



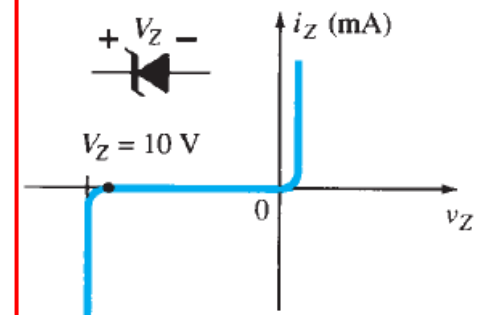
$$V_L = V_Z = 10 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 10 \text{ V} = 6 \text{ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{10 \text{ V}}{3 \text{ k}\Omega} = 3.33 \text{ mA}$$

$$I_R = \frac{V_R}{R} = \frac{6 \text{ V}}{1 \text{ k}\Omega} = 6 \text{ mA}$$

$$\begin{aligned} I_Z &= I_R - I_L \\ &= 6 \text{ mA} - 3.33 \text{ mA} \\ &= 2.67 \text{ mA} \end{aligned}$$



Problem:2

For the Zener diode network of Fig. below , determine V_L , V_R , I_Z , and P_Z .

